

SurfaTech Invoice Automation System

Project Report & Technical Architecture

A Google Forms + Google Sheets + Google Apps Script based workflow for capturing customer and product details, generating structured invoice numbers, calculating invoice values, and preparing invoices for automated output.

Organization	SurfaTech Integrated Solutions
Invoice format	SIS/2026-27/0023 (example)
Current financial year	2026-27
Automation platform	Google Workspace + Google Apps Script

1. Project Overview

The project is designed to reduce manual invoice preparation by collecting customer and product information through a Google Form and processing the submitted response in Google Sheets. Google Apps Script provides the automation layer, including invoice-number generation and form-submit event handling.

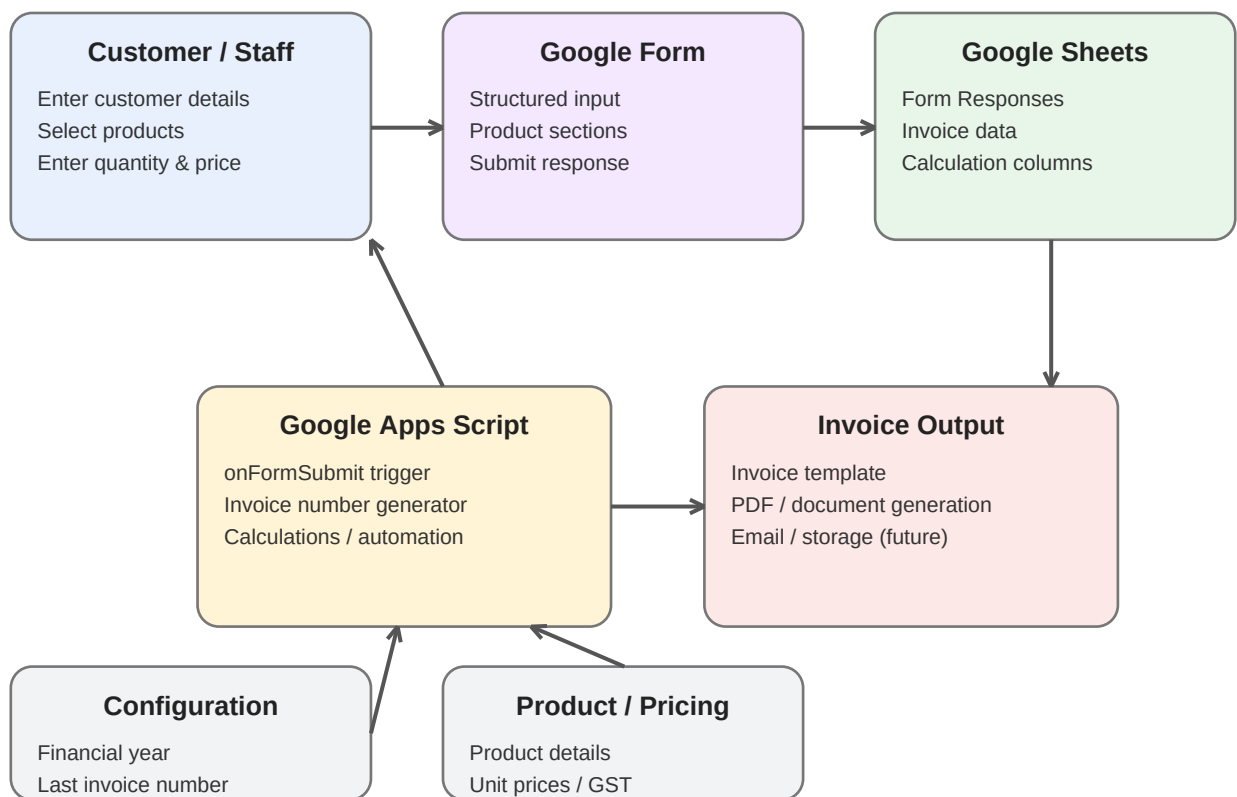
Core workflow

- Customer or staff member submits the Google Form.
- The response is stored automatically in the linked Google Sheet.
- Apps Script receives the form-submit event.
- The script generates the next invoice number using the financial-year prefix and four-digit sequence.
- Product quantity, unit price, amount, subtotal and GST-related fields can be calculated.
- The processed data can then be used to generate an invoice document/PDF.

2. System Architecture

The architecture separates data collection, data storage/calculation, automation logic, and invoice output. This makes the system easy to test incrementally and allows invoice generation to be expanded later without changing the customer-facing form.

SurfaTech Invoice Automation – System Architecture



Form submission → process data → invoice number → calculations → invoice output

Architecture components

Component	Purpose
Google Form	Captures customer details, GSTIN, product names, quantities and unit prices.
Form Responses Sheet	Stores each submission as a row and acts as the initial data source.
Google Apps Script	Runs automation when a form is submitted; handles invoice numbering and processing.
Configuration	Stores financial-year information and the latest invoice sequence.
Invoice Output	Uses processed data to create an invoice document/PDF; email/storage can be added later.

3. Google Form & Spreadsheet Structure

The form response sheet used during implementation contains customer fields followed by repeated product blocks. The product blocks support multiple products on a single invoice.

Area	Example fields
Customer	Customer Name, Company Name, Billing Address, Phone Number, Email, GSTIN
Product 1	Product Name - 1, Quantity - 1, Unit Price - 1, Amount - 1
Product 2	Product Name - 2, Quantity - 2, Unit Price - 2, Amount - 2
Product 3–5	Additional product, quantity, unit price and amount fields
Totals	Subtotal, GST %, GST Amount, Grand Total

4. Invoice Number Generation

The required invoice format is **SIS/2026-27/0023**. The financial year identifies the current accounting year, while the final four digits represent the sequential invoice number.

- **Prefix:** SIS
- **Financial year:** 2026-27
- **Sequence:** 0023, 0024, 0025, ...
- **Next invoice after 0023:** SIS/2026-27/0024

During testing, the Apps Script function produced SIS/2026-27/0024 from a last invoice sequence of 23. For production use, the sequence should be stored centrally rather than hard-coded so concurrent submissions do not produce duplicate invoice numbers.

5. Google Apps Script Automation

Recommended production flow

- Receive the submitted row from the onFormSubmit event.
- Read the current financial year and last invoice sequence.
- Increment the sequence and format it with four digits using leading zeroes.
- Write the invoice number into the processed row.
- Calculate each product amount as Quantity × Unit Price.
- Calculate subtotal, GST amount and grand total.
- Populate an invoice template with customer and line-item data.
- Export/store the invoice as PDF and optionally email it to the customer.

6. Testing & Reset Procedure

A test submission was used to verify the form-to-sheet workflow and the invoice-number function. Once testing is complete, test rows should be removed or clearly marked as test data before production use. The invoice sequence should also be reset to the intended production starting point.

Production checklist

- Remove test customer records from the response/processing sheet.
- Confirm the starting invoice number for the current financial year.
- Confirm the financial-year value changes automatically at the beginning of the new year.
- Install and verify the spreadsheet on-form-submit trigger.
- Submit one controlled test invoice and verify all calculations.
- Verify the generated PDF/document before using the system for live invoices.

7. Security & Reliability Considerations

- Restrict spreadsheet editing permissions to authorized staff.
- Do not expose the response spreadsheet publicly because it contains customer and GST-related information.
- Use Apps Script locking or another atomic sequence mechanism to avoid duplicate invoice numbers when two submissions occur simultaneously.
- Keep the invoice template and generated files in a controlled Google Drive folder.
- Maintain backups/export copies of finalized invoices and important configuration.

8. Future Enhancements

- Automatic PDF invoice generation from a professional invoice template.
- Automatic email delivery to the customer.
- Product master sheet with predefined product names, prices and GST rates.
- Automatic financial-year rollover.
- Invoice search and dashboard.
- Invoice status tracking such as Draft, Generated, Sent and Paid.
- Centralized invoice-number counter with concurrency protection.
- Audit log for generated and modified invoices.

9. Conclusion

The SurfaTech Invoice Automation System provides a practical low-cost foundation for digitizing invoice preparation. The current implementation establishes the core data-collection and automation pipeline. The next major milestone is connecting the processed spreadsheet row to a standardized invoice template and automatically producing a final PDF.